Syllabus
SURV 615 (UMD)/SURVMETH 685 (UMICH)
Statistical Methods and Machine Learning I
Fall 2024

(Last updated: August 27, 2024)

Instructor	John Kubale (jkubale@umich.edu)
Meeting Time:	Tuesdays, 9:30AM-12:00PM
Meeting place:	1070 ISR-Thompson (UMICH) 1208 Lefrak (UMD)
Office hours:	Fridays 2:00-4:00 PM EDT (Via Zoom) – starting Sept. 6
TA:	Weiyushi (Sarah) Tian (wystian@umich.edu)
Class Website:	All the course-related material will be posted on Canvas: https://umich.instructure.com/courses/702278 If you have any questions regarding Canvas or Zoom access, please contact Elisabeth Schneider (dodo@umich.edu). This term we will be using Piazza for class discussion. Please use the following class sign-up link: https://piazza.com/umich/fall2024/survmeth685001fa2024/info
Video Site:	Zoom will also be accessible on Canvas. All lectures will be recorded and made available through Zoom/Canvas. Class attendance is part of the grade; the video recordings should be used for emergency situations or reviews not as main means of learning.
Textbooks/Materials:	1. Faraway, J. J. (2014). Linear models with R. 2nd Edition. CRC press. 2. James, G., Witten, D., Hastie, T., and Tibshirani, R. (2021). An introduction to statistical learning. 2nd Edition. New York: Springer. (Noted as JWHT. Downloadable at https://www.statlearning.com/) 3. All other materials will be provided via Canvas.
Software:	R See Faraway Appendix A and JWHT Ch 2.3. Come to Class 1 with the following installed on your computer: 1. R: See https://www.r-project.org/ 2. R packages MASS, faraway, ggplot2, dplyr, and ISLR 3. RStudio: See https://www.rstudio.com/

Overview

This is the first course in a three-term sequence in applied statistical methods and machine learning that are the basis in handling complex datasets. The purpose of this class is to learn basic statistical methods through the use of linear model theory and regression. The emphasis is to understand data, apply the methods presented and develop a feel for how problems in data analysis can be viewed from different perspectives. In all cases, the focus will be on understanding the proper techniques, rather than deriving their theoretical properties. The students are expected to increase their understandings about the techniques through quizzes, lab sessions, projects, mid-term exam and final exam. R programming is necessary for this class, and this class is designed to offer basic programming instructions suitable for the content of this class through lab sessions during the last portion of each class. The students are expected to have basic R skills and make an improvement throughout the course with examples given from the lectures, lab sessions, and homeworks.

Outline

Class 1 (Aug 27)
Quantitative research; Introduction to R and R Studio
Reading: Faraway Ch 1.2, Appendix A; JWHT Ch 1, 2.3

Class 2 (Sept 3)
Parameters; Variance; Hypothesis testing
Reading: Old class notes
Homework #1 assigned

Class 3 (Sept 10)
Quiz #1
Statistical learning; When to use linear models; Estimation; hypothesis tests as linear models
Reading: Faraway Ch 1.3, 1.4, 2.1, 2.2; JWHT Ch 2.1, 2.2
Homework #1 due; Homework #2 assigned

Class 4 (Sept 17)
Quiz #2
Simple regression; Ordinary least square (OLS); Assessing models
Reading: Faraway Ch 2.3-2.5; JWHT Ch 3.1
Homework #2 due; Homework #3 assigned

Class 5 (Sept 24)
Quiz #3
DAGS; Multiple regression; Inference; Goodness of fit;
Reading: Faraway Ch 2.9,3; JWHT Ch 3.2
Homework #3 due; Homework #4 assigned

Class 6 (Oct 1)
Exam 1: (Covers materials from classes 1-5)
Homework #4 due

Class 7 (Oct 8)
Quiz #4
Categorical predictors; Interactions
Reading: Faraway Ch 14; JWHT Ch 3.3.1, 3.6.4
Homework #5 assigned

Fall Break – No class October 15

Class 8 (Oct 22)
Quiz #5
Prediction vs. Explanation
Reading: Faraway Ch 4-5
Homework #5 due; Homework #6 assigned

Class 9 (Oct 29)
Quiz #6

Model diagnostics
Reading: Faraway Ch6; JWHT Ch 3.3.3
Homework #6 due; Homework #7 assigned

Class 10 (Nov 5)
Exam 2: (Covers material from Classes 1-8)
Homework #7 due

Class 11 (Nov 12)
Quiz #7
Problems with predictors and errors; Transformation
Reading: Faraway Ch 7-9; JWHT Ch 3.3.3
Homework #8 assigned

Class 12 (Nov 19
Quiz #8
Model selection
Reading: Faraway Ch 10; JWHT Ch 6.1, 6.5
Homework #8 due; Homework #9 assigned

Class 13 (Nov 26)
Quiz #9
Study designs – Experimental design and observational studies
Reading: Faraway Ch 15-16
Homework #9 due; Homework #10 assigned

Class 14 (Dec 3)
Quiz #10
Resampling methods; Cross-validation; Bootstrap
Reading: JWHT Ch 5.1, 5.2
Homework #10 due

Class 15 (Dec 10)
Exam 3: Covers materials from Classes 1-14

Class Policies

Exams:

- There will be 3 exams (all are cumulative)
- **All work on exams must be your own. Do not discuss the exam with others.**

Quizzes:

- To be taken on Canvas at the beginning of each class (starting from Class #3)
- Timed at 5 minutes
- 3 questions in true/false, multiple choice, or select-all-that-apply formats on topics covered in previous class
- Answers will be reviewed immediately
- Will be graded on Canvas

Lectures:

- Will start right after quiz session

- Lecture notes will be made available on Canvas prior to each class

Lab sessions:
- Roughly speaking, will be last 30 minutes of each class
- Code and output will be made available prior to each class
- Will focus on the implementation of lecture materials using R
- Will be structured to help complete homeworks

Homeworks:
- Will be grouped with another student (possibly 2 in rare cases) to work with throughout term
- The project assignment and due dates are provided in outline (see above)
    - Assigned homeworks will be made available on Canvas at 9:30AM on day they are assigned
    - Homeworks are due by 9:30AM of due date
    - Late submissions will be accepted only on permission and will generally receive 50% of the points unless discussed
- Submission guides:
    - Prepare in pdf format. No handwritten submissions
    - Include your names and the homework number at the top of the first page
    - One project submission per group
    - Submit via Canvas
- Will be graded on Canvas

Use of AI/LLMs:

Use of large language models like ChatGPT, UMICH GPT, etc. and other AI tools (e.g., Github copilot) is **not allowed on any of the exams or quizzes** and is in general strongly discouraged in this class (see below). **If you do choose to use them on a homework you must cite everywhere it was used.** If you are found to have used such tools on an exam or quiz, or on a homework without citing each instance, this will be considered academic misconduct.

COVID-19 and illness statement:

For the safety of all students, faculty, and staff on campus, it is important for everyone to comply with safety measures that have been put in place for our protection. We each have a responsibility for protecting the collective health of our community. Your participation in this course on an in-person basis is conditional upon your adherence to all safety measures mandated by the University of Michigan or the University of Maryland. Both universities have a vaccination policy: https://ehs.umich.edu/wp-content/uploads/2021/07/COVID-19_Vaccination_Policy.pdf and https://umd.edu/4Maryland#vaccines. As of August 16, 2022, masks are not required while indoors with a few exceptions but strongly recommended at both universities. However, this may change. **Individuals are expected to stay home if they are sick**. This helps reduce the likelihood of spreading a range of infections including COVID-19, influenza and other illnesses.

Academic conduct:
- It is extremely important to familiarize yourself with respective campus policies on academic integrity.
- UMD: https://policies.umd.edu/assets/section-iii/III-100A.pdf
- UMICH: https://rackham.umich.edu/academic-policies/section8/

Accommodations:
- UMD: In order to receive services you must contact the Disability Support Services

(DSS) office to register in person for services. Please call the office to set up an appointment to register with a counselor. Contact the Accessibility & Disability Service; http://www.counseling.umd.edu/ads/

- UMICH: If you think you need an accommodation for a disability, please contact Services for Students with Disabilities (SSD) office to help us determine appropriate academic accommodations; http://ssd.umich.edu. Any information you provide is private and confidential and will be treated as such.
- Students who expect to miss classes, examinations, or other assignments as a consequence of their religious observance shall be provided with a reasonable alternative opportunity to complete such academic responsibilities. It is the obligation of students to provide faculty with reasonable notice of the dates of religious holidays on which they will be absent.
- Please visit http://faculty.umd.edu/teach/attend_student.html and http://www.provost.umich.edu/calendar/ for the complete University policy at the respective institutions.

Grading

Quizzes (10%)
Homework (25%)
Exam 1 (20%)
Exam 2 (20%)
Exam 3 (20%)
Class Participation (5%)

The final course grade is determined by assigning the letter grade corresponding to the result of the weighted average of the numerical scores for each of the assignments as follows.

A+	[98, 100]
A	[93, 98)
A-	[90, 93)
B+	[87, 90)
B	[83, 87)
B-	[80, 83)
C+	[77, 80)
C	[73, 77)
C-	[70, 73)
D+	[67, 70)
D	[63, 67)
D-	[60, 63)
F	<60